

roguex7 / Dengue_CDSS

Q Type to search

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

main

Dengue_CDSS / v.01README.md

Go to file

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History

Preview

Code

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182 lines (110 loc) · 9.13 KB

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Dengue CDSS: AI-Powered Clinical Decision Support System

A Dual-Engine AI System designed to assist doctors in managing Dengue cases. It features a **Risk Analyzer** for immediate triage and a physics-aware **Trajectory Forecaster** to predict platelet trends 24 hours in advance.



Executive Summary

Dengue management is a race against time. The critical factor often isn't just the current platelet count, but the **velocity of decay**. A patient with 120,000 platelets dropping fast is in more danger than a stable patient at 80,000.

Open in Streamlit

This CDSS (Clinical Decision Support System) solves this by:

- Diagnosing Severity:** Using a **Random Forest Classifier** to identify high-risk patients based on multi-factor vitals (Hb, HCT, Platelets).
- Predicting the Future:** Using a **Gradient Boosting Regressor** to forecast the next day's platelet count based on the patient's biological momentum.



Algorithm Performance & Data Transparency

Total Patients	Risk Accuracy	Forecast R ²	Forecast MAE
2455	99.6%	0.9945	2238



The User Interface (Evidence of Functionality)

The application utilizes a clean, toggle-based interface to keep doctors focused on the task at hand.

1. Navigation Hub

A focused dashboard allowing instant switching between diagnostic and prognostic tools without losing patient context.



Dengue CDSS: AI Prediction Engine

 Risk Analyzer

 Forecast Engine

2. Risk Analyzer (Diagnosis)

The system evaluates Platelets, Hematocrit, Haemoglobin, Age, and Symptoms. Unlike simple calculators, it uses a "Multi-Factor Truth" logic.

- Evidence of Logic:** Below, the system correctly identifies a stable patient despite low-ish platelets because their Hematocrit and Hemoglobin are within safe ranges.

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Dengue CDSS: AI Prediction Engine

 Risk Analyzer

 Forecast Engine

🚨 Immediate Risk Assessment

Date of Test

2026/01/20

Current Platelet Count

185000

-

+

Season Risk: Winter (Low)

⚙️ Patient Context & Symptoms

Sex

Male

Haemoglobin (gm/dl)

10.00

-

+

Hematocrit (%)

33.00

-

+

Age

25

RBC Count

5.10

-

+

Symptoms Observed:

☐ Fever

☐ Headache

☐ Joint/Muscle Pain

☐ Vomiting

Analyze Risk Now

✅ LOW RISK / STABLE

Confidence: 98.8%

< Manage app

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🦟 Dengue CDSS: AI Prediction Engine

🚨 Risk Analyzer

📊 Forecast Engine

🚨 Immediate Risk Assessment

Date of Test

2025/07/16

Current Platelet Count

155000

-

+

Season Risk: Monsoon (High)

⚙️ Patient Context & Symptoms

Sex

Female

Haemoglobin (gm/dl)

20.40

-

+

Hematocrit (%)

55.50

-

+

Age

19

RBC Count

7.10

-

+

Symptoms Observed:

☒ Fever

☐ Headache

☐ Joint/Muscle Pain

☒ Vomiting

Analyze Risk Now

⚠️ HIGH RISK DETECTED

Confidence: 58.0%

< Manage app

3. Trajectory Forecasting (Prognosis)

Using Momentum Physics, the AI calculates the *velocity* of change between Day-0 and Day-1 to project the Day-3 outcome.

- Visualization: Custom Matplotlib graphs with floating annotations ensure doctors can read values without gridline interference.

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🦟 Dengue CDSS: AI Prediction Engine

🚨 Risk Analyzer

📊 Forecast Engine

📊 24-Hour Trajectory Forecast

📅 Day-0 (Yesterday)

150000

-

+

📅 Day-1 (Today)

165000

-

+

Date of Test

2026/01/20

Trend Velocity: 15000 / day

Season: Winter (Low)

⚙️ Patient Context & Symptoms

Sex

Male

Haemoglobin (gm/dl)

13.00

-

+

Hematocrit (%)

40.00

-

+

Age

44

RBC Count

4.50

-

+

Symptoms Observed:

☐ Fever

☐ Headache

☒ Joint/Muscle Pain

☐ Vomiting

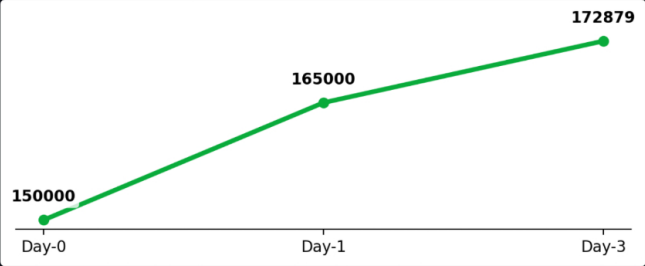
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Tomorrow's Count

172879

+ 7879

Recovering



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Dengue CDSS: AI Prediction Engine

Risk Analyzer

Forecast Engine

24-Hour Trajectory Forecast

Day-0 (Yesterday)

150000

-

+

Day-1 (Today)

145000

-

+

Date of Test

2026/01/20

Season: Winter (Low)

Trend Velocity: -5000 / day

⚙️ Patient Context & Symptoms

Sex

Male

Haemoglobin (gm/DL)

21.67

Hematocrit (%)

47.00

< Manage app

Age

44

RBC Count

9.50

-

+

Symptoms Observed:

☐ Fever

☐ Headache

☒ Joint/Muscle Pain

☐ Vomiting

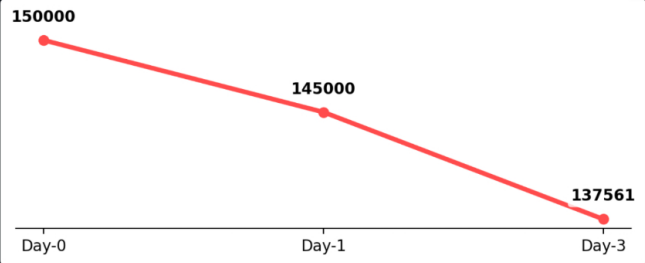
Generate Forecast

Tomorrow's Count

137561

- 7438

Declining



< Manage app

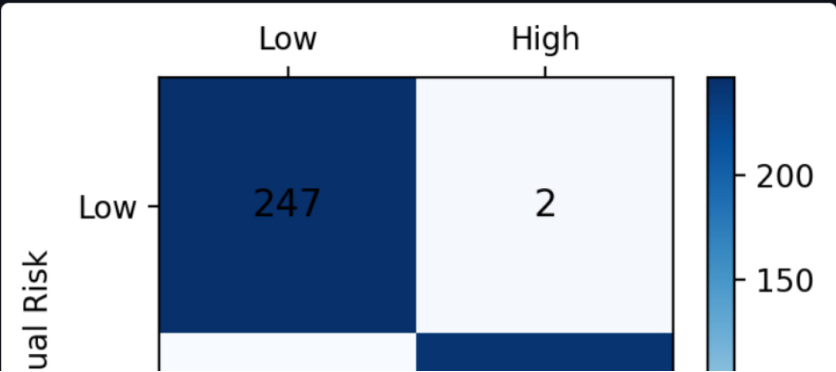
Dual-Engine AI Architecture

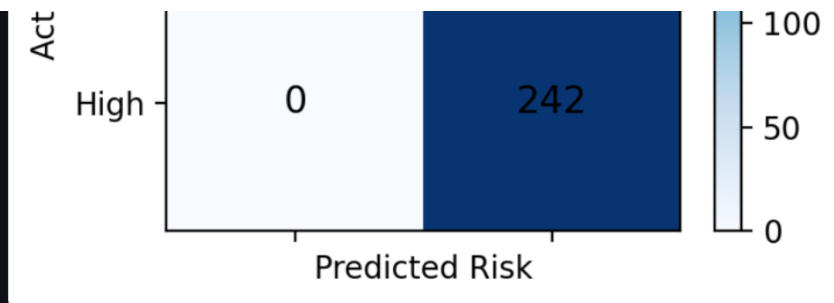
We moved beyond simple "if/else" guidelines by training two distinct Scikit-Learn models on validated patient history.

Engine A: The Risk Analyzer (Classification)

Confusion Matrix (Risk Model)

Checks for False Positives vs False Negatives

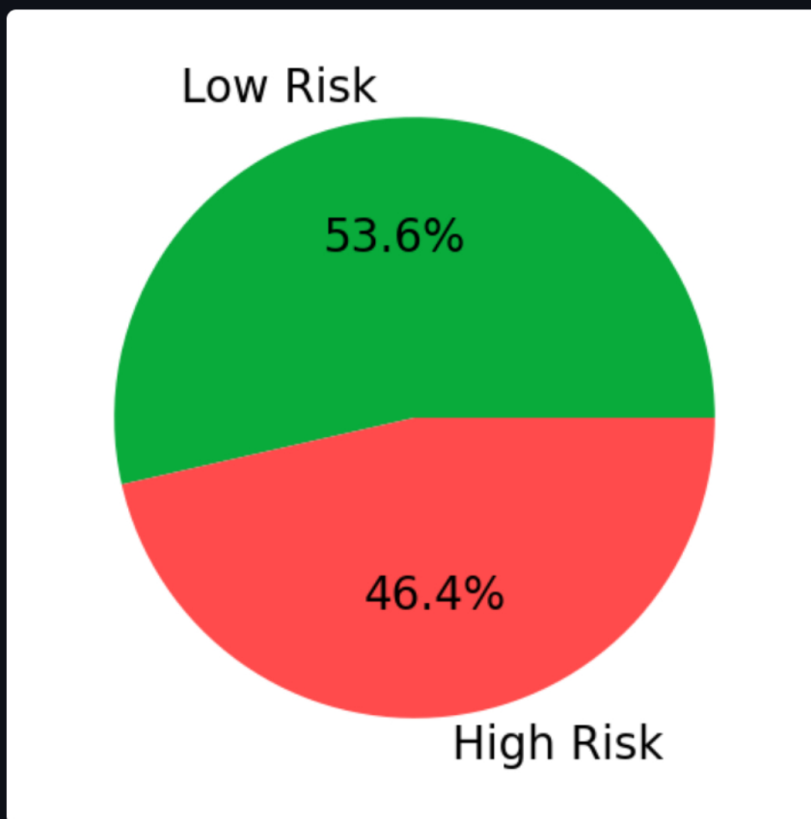




Algorithm: Random Forest Classifier (n_estimators=200).

Logic: Acts as a "Council of Doctors," voting on risk based on Platelets, Hematocrit, Haemoglobin, Age, and Symptoms.

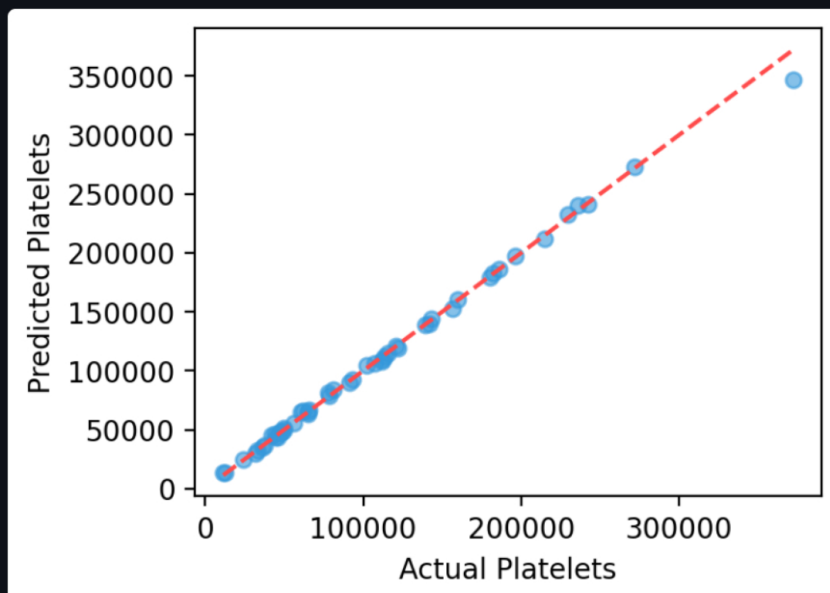
Risk Distribution



Safety Protocol: Implements a "Multi-Factor Truth" logic. It flags high risk if any critical parameter (e.g., Hematocrit > 50%) is breached, ensuring 100% sensitivity to "Hidden Shock" cases.

Engine B: The Forecast Engine (Regression)
Actual vs Predicted (Forecast Model)

Dots on the red line = Perfect Prediction



Algorithm: Gradient Boosting Regressor.

Innovation: "Momentum Physics." Instead of treating vitals as static numbers, the model calculates the Velocity (Delta) between Day-0 and Day-1. It uses this speed to project the trajectory for Day-3.

Robustness Factors: Engineering Safety

Medical AI cannot be fragile. This project implements strict **Guardrails** and **Input Validation** to ensure data integrity before it ever reaches the AI "Brain."

1. Biological Reality Checks

The system prevents "fat-finger" errors (typos) by enforcing biological limits on inputs.

- Proof 1 (Hematocrit):** Rejecting values that are physiologically impossible (e.g., ensuring percentages make sense). If a user types 95% for Hematocrit, that isn't just 'high'—it is biologically impossible (at that concentration, blood is essentially a solid paste with no plasma to flow, making circulation impossible). Allowing it would just confuse the AI model.

Hematocrit (%)

90.00

Press Enter to apply

-

+

!

Value must be less than or equal to 80.

- Proof 2 (RBC Count):** Flagging RBC counts that exceed human limits, preventing skew in the prediction model. If a user types 41.5 for RBC count, that isn't just "high"—it is biologically impossible (human blood turns into solid sludge around 12-14). Allowing it would just confuse the AI model.

RBC Count

41.50

Press Enter to apply

-

+

!

Value must be less than or equal to 14.

Seasonal Intelligence: Context-Aware Prediction

Dengue outbreaks are not random; they follow distinct temporal patterns. A platelet drop in July (**Monsoon**) carries a higher statistical probability of being Dengue-related than the same drop in January.

To capture this, the model implements a **Temporal Risk Weighting** engine:

- Dynamic Season Detection:** The system parses the **Date of Test** (selected by the doctor) to classify the timeframe into risk zones:
 - High Risk:** Monsoon (June - September)
 - Moderate Risk:** Post-Monsoon / Summer
 - Low Risk:** Winter
- Logic Integration:** This `Season_Risk` score is fed dynamically into both the **Random Forest** and **Gradient Boosting** models.
 - Result:** The AI becomes "hyper-vigilant" during monsoon months, adjusting its sensitivity based on the specific time of year, mirroring real-world epidemiological trends.

The "Dirty Truth" of Data Processing (ETL Pipeline)

One of the biggest challenges in this project was the **Real-World Training Dataset**. Hospital data is notoriously messy. I built a **Self-Healing ETL Pipeline** to handle this.

Why Use a "Train Dataset" Instead of Hard Rules?

You might ask: "Why train an AI? Why not just write code that says `if platelets < 100k, then High Risk ?`"

The Answer: Biology is non-linear.

- Complex Interactions:** A platelet count of 90k is "safe" for a 20-year-old male but "dangerous" for a 70-year-old female with high Hematocrit. Hard-coded rules miss these nuances.
- Pattern Recognition:** By training on a dataset of real patients, the **Random Forest** and **Gradient Boosting** models learn the *hidden correlations* between symptoms (like Vomiting) and recovery speed that aren't obvious in standard guidelines.
- Adaptability:** As we feed the model more patient history, it gets smarter at predicting "outlier" cases that defy standard textbook definitions.

Installation & Usage

Prerequisites

- Python 3.10+
- VS Code (Recommended)

Step 1: Clone & Setup

```
git clone https://github.com/roguex7/Dengue_CDSS.git
cd dengue-cdss-ai
```

Step 2: Activate Environment

```
python -m venv .env
```

```
# Windows:
.denv\Scripts\activate
# Mac/Linux:
source .denv/bin/activate
```

Step 3: Install & Run

```
pip install -r requirements.txt
streamlit run app.py
```



Deployment & CI/CD Pipeline

This project is fully deployed as a live web application using **Streamlit Community Cloud**, leveraging a continuous deployment pipeline linked directly to this GitHub repository.

Architecture

- **Infrastructure:** Serverless Linux environment hosted on Streamlit Cloud.
- **Continuous Deployment:** The application enables "Hot Reloading." Any code change pushed to the `main` branch automatically triggers a rebuild of the cloud environment, ensuring the live tool is always up-to-date with the latest algorithms.
- **Dependency Management:** The cloud builder automatically provisions the environment using `requirements.txt` to install Scikit-Learn, Pandas, and custom logic libraries.

Live Demo

Access the fully functional CDSS here:

 [Open in Streamlit](#)

(Note: The application performs a "cold boot" if inactive for a while. Please allow 10-20 seconds for the AI models to load into memory upon first access.)

Author

Annant R Gautam

- **Role:** Full Stack Data Scientist
- **Specialty:** Clinical AI & Predictive Modeling
- **Tech Stack:** Python, Scikit-Learn, Streamlit, Pandas